DMCAST: A Prediction Model for Grape Downy Mildew Development

DMCAST: Ein Modell zur Vorhersage der Entwicklung von Falschem Mehltau auf Reben

DMCAST: un modèle de prédiction du dévelopement du mildiou de la vigne

Eun Woo Park, Robert C. Seem, David M. Gadoury, and Roger C. Pearson

Visiting scientist, associate professor, senior research associate, and professor (deceased), respectively.
Department of Plant Pathology, Cornell University, New York State Agricultural Experiment Station, Geneva, 14456, USA.
Present address of first author: Department of Agricultural Biology, Seoul National University, Suwon 441-744, South Korea.

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Abstract
A weather-driven prediction model (DMCAST) that simulated the processes of oospore maturation, primary infection, and secondary infection was developed and evaluated for managing grape downy mildew epidemics in New York. The proportion of oospores that matured each day was estimated based upon daily precipitation from 21 September the previous year. When more than 3% of the oospores were mature, and five to six leaves of the host had unfolded, the date of primary infection was predicted when weather conditions were suitable for primary infection. After primary infections, DMCAST estimated the incubation period based upon hourly temperature, and then simulated sporulation, spore mortality, spore survival, and secondary infection based upon hourly temperature, relative humidity, surface wetness, and hours of darkness. Observation of oospore maturation in Geneva and Riverhead, NY suggested the model was able to simulate oospore maturation in Geneva, but required modification for use in other locations. DMCAST prediction of the date of primary infection and initial sporulation appeared to be accurate when compared with the date of initial downy mildew symptoms and seasonal disease severity in 1985-1993. Over a 3-yr period, disease incidence increased significantly following most infection periods predicted by DMCAST. However, there was no increase in disease incidence after 1 of 11, 6 of 19, and 1 of 5 infection periods in 1990, 1992, and 1993, respectively. In no case did disease increase in the absence of infection periods predicted by DMCAST, *i.e.*, when conditions were predicted to be unfavorable for epidemic development. Post-infection applications of metalaxyl timed according to DMCAST infection periods significantly and substantially reduced the severity of fruit infection from that observed on unsprayed vines in 1992 and 1993. Under severe disease pressure in 1992, 1, 2, or 3 post-infection sprays reduced disease severity to 12.6, 9.4, or 9.3%, as compared to 50.4% on unsprayed vines, and 0.1% on vines that received seven protectant sprays. Under less favorable conditions for epidemic development in 1993, disease control on vines that received a single post-infection spray was equal to that obtained through 6 protectant applications. DMCAST appears to be an accurate predictor of primary infection, and an imperfect, but useful prediction model for secondary infection periods.
Key Words: epidemiology, disease forecasting, disease management, simulation, *Plasmopara viticola*

1. Introduction

Downy mildew, caused by the fungus *Plasmopara viticola* (Berk. & Curt.) Berl. & de Toni, is potentially one of the most damaging diseases of grape in the world (Pearson and Goheen 1988). However, annual variations in the complex and peculiar weather conditions favorable to disease development often result in unexpected outbreaks of the disease, and expected outbreaks that never occur. Consequently, growers may resort to the application of several protectant fungicide sprays (with or without benefit), or may risk serious crop loss by withholding fungicides when they are indeed necessary. For example, the wet growing season in 1986 demonstrated to many growers in the Finger Lakes and Lake Erie regions of New York the inadequacy of their downy mildew control programs during the wet summer. An accurate prediction system for downy mildew might have substantially reduced losses due to downy mildew in 1986 by providing a timely warning of the impending epidemic.

Significant progress has been made in the United States, Australia, and Europe with regards to empirical and/or mathematical models to predict downy mildew infection (Kable 1977; Kable *et al* 1990; Lalancette *et al* 1988; Lalancette *et al* 1988; Magarey 1990;Strizyk 1983; Tran Manh Sung 1987; Tran Manh Sung *et al* 1990). For example, the EPI model, which was developed by Strizyk (1983) in France, identified general weather conditions that were favorable for the fungal development by comparing current weather conditions to long-term average climatological conditions. Although the EPI model has been validated in the Bordeaux area (Maurin 1983), it's empirical nature makes it difficult to adapt to other regions. In a field trial conducted in Geneva in 1990, EPI was not a sufficiently accurate predictor of downy mildew (Seem unpublished).

Recent progress in computer-aided technologies for weather monitoring and data processing, mathematical models have great potential to improve practical use of disease forecasting systems. As for grape downy mildew, mathematical models have been derived from biological studies on the effects of environmental conditions on various phases of the infection cycle of *P. viticola* (Blaeser and Weltzien 1978; Blaeser and Weltzien 1979; Gehmann *et al* 1987; Lalancette *et al* 1988; Lalancette *et al* 1988; Leu and Wu 1982; Muller and Rabanus 1923). These models usually predict sporangium formation, infection periods, and incubation only based upon environmental conditions assuming that enough inoculum is initially available. This assumption can result in the prediction of infection periods when initial inoculum is not available.

The present study was undertaken to improve capability of predicting risk levels of downy mildew infection by logically integrating

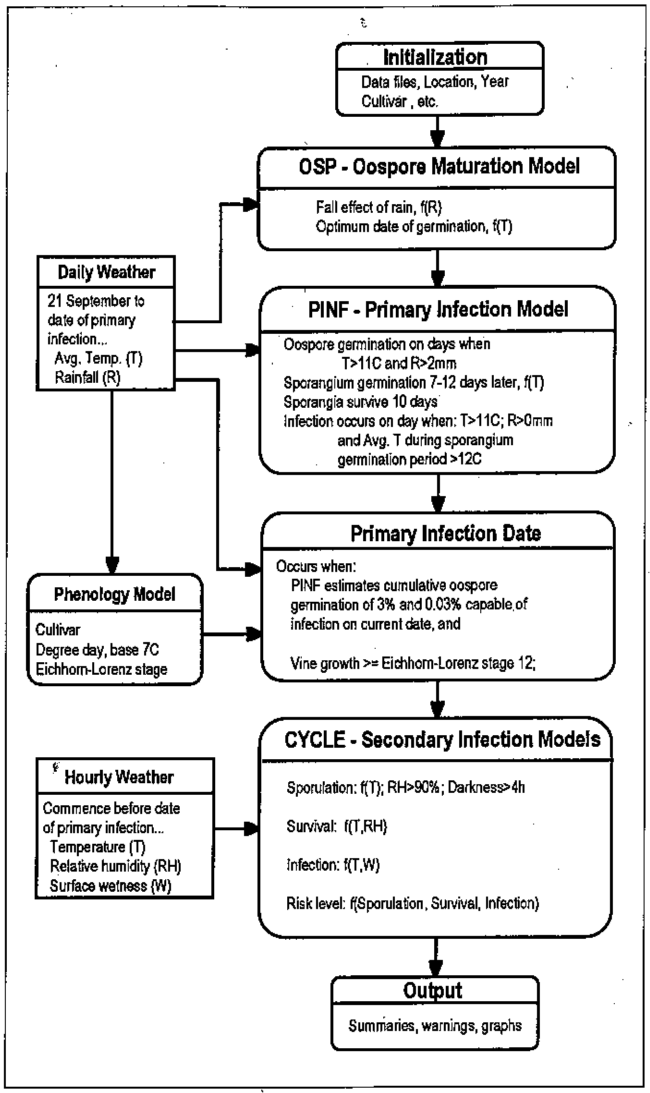

Fig. 1. Simplified flowchart of DMCAST showing data input requirements and basic processes used for oospore maturation, primary infection, and secondary infection.

Abb.1: Vereinfachtes Fließschema von DMCAST, das die erforderlichen Daten und Prozesse zeigt, die notwendig sind für die Oosporenreifung, Primär- und Sekundärinfektion

Fig. 1: Explication simplifiée du système DMCAST montrant les données nécessaires À introduire et les procédés de bases utilisés pour la maturation des oospores, l'infection primaire et l'infection secondaire.

empirical and experimental knowledge of overwintering and primary infection processes and secondary infection cycles of *P. viticola*. A weather-driven prediction model which was named DMCAST (Downy Mildew foreCAST model) was developed in this study. Validity of the model was evaluated using disease and weather data collected in 1985 through 1993.

2. Model Description

DMCAST is designed to provide a daily estimate of the level of oospore maturation from 1 January, to determine the date of primary infection, and to predict the occurrence and severity of secondary infection periods (Fig. 1). DMCAST was programmed in both Turbo C (Borland, Inc.) and Fortran-77 (Microsoft Corp.), and can be run in all DOS-based computers. Daily average temperature and rainfall from 21 September of the previous year the first primary infection, and hourly data on temperature, relative humidity, wetness period during the growing season (usually commencing on 1 May) are necessary to run the model. Source code for the model, sample data files, and the executable program are available from the second author.

2.1. Oospore maturation.

The oospore maturation (OSP) model estimates the distribution of oospore maturation from 1 January as described by Tran Manh Sung (1987). According to her investigation, oospore maturation appeared to be associated with monthly rainfall during the overwintering period, and the dynamics of oospore maturation was represented by a bell-shaped curve similar to the normal distribution curve (Richards 1959; Tran Manh Sung 1987; Tran Manh Sung *et al* 1990). We estimated the probability of oospore maturation on each day with the normal distribution density function:

$$P(k) = 1/(v\bullet SQRT(2\bullet p))\bullet EXP(-0.5\bullet((k\bullet D)/v)^2)$$

where $P(k)$ is the probability of oospore maturation on k^{th} day from 1 January, D and v are respectively the mean and standard deviation of the number of days from 1 January that is required for oospores to mature *pi* is the mathematical constant 3.1416, and SQRT and EXP are the square root and exponentiation functions ,respectively. D and v were determined with the cumulative value of rain effect (Ra) calculated at the end of January by the following equations which were described by Tran Manh Sung (1990) and parameterized by Tran Manh Sung and Seem (unpublished).

$$D = 118 - 0.3Ra$$

$$v = 13.5 + 0.02Ra$$

Therefore, the number of days required for oospores to mature was distributed normally with the mean of D and the standard deviation of v. The oospore maturation model of DMCAST regarded the daily probability of oospore maturation as the proportion of mature oospores in the whole, seasonal population of oospores on each day.

2.2 Primary infection.

The primary infection (PINF) model of DMCAST determines the dates of primary infection by the overwintered inoculum using daily temperature and rainfall data. Parameters representing effects of temperature and rainfall on the stages of infection process were adopted from the PCOP model, which was developed by Tran Manh Sung (1987) to predict the primary infection date from the date when most oospores are mature (DOM).

In the primary infection model, the cohorts of oospores which matured on each day after 1 January were considered capable fulfilling the infection processes (oospore germination, sporangium formation, and vine infection), subject to the influence of daily temperature and rainfall. Oospore germination occurred on days when the average temperature was above 11° C and rainfall exceeded 2 mm. Sporangium germination took place 7-12 days later; the actual time period was a function of average daily temperature. Primary infection could occur on subsequent days when any rainfall occurred and the average temperature was greater than 11° C. Since each cohort of mature oospores was subjected to different environmental conditions after the date of maturation, the number of days needed by individual cohorts to achieve primary infection differed from each other. Consequently, the model allowed cohorts of oospores matured on different dates to infect on the same date. For each date of infection, proportions of oospores which were estimated for each cohort in the oospore maturation model were accumulated to determine the proportion of total oospores that could infect vines on that day. However, if more than 10 days were taken to reach infection after sporangium formation, the inoculum was arbitrarily considered to be no longer viable and was not included in the accumulation of oospores contributing to vine infection.

The date of primary infection was finally predicted based on two additional assumptions. First, primary infection occurred only after 3% of the overwintered oospores had matured. Second, vine growth must reach Eichhorn-Lorenz (Pearson and Goheen 1988) growth stage 12, (i.e., 5-6 leaves unfolded) before infection can occur. The first assumption was based upon the relationship between predicted oospore maturity and observed disease in New York. When the primary infection model was run with various threshold levels of matured oospores using weather data from 1985 to 1992, the 3% level provided expected primary infection dates which best approximated the first observation dates of downy mildew symptoms. The second assumption was based on observed dates of initial downy mildew symptoms from 1985-1989, and the stage of vine growth at the probable date of infection. A phenology model to estimate the occurrence of the vine growth stages was developed for 23 grapevine cultivars, and were incorporated in the primary infection model (Park, unpublished). The phenology model estimated vine growth in terms of the Eichhorn-Lorenz index and based the development on degree days of daily maximum temperatures from 1 January using 7° C as the base. The Richards growth function (Richards 1959) was fitted to vine growth data collected in Geneva to develop the phenology models.

2.3 Secondary infection cycles.

Once primary infection was expected, the secondary infection cycle model (CYCLE, Fig. 1) estimated the incubation period of *P. viticola* based upon hourly temperatures. CYCLE then calculated hourly factors for sporulation (0-1), spore mortality (0-1), spore survival (0-1), and environmental favorability for infection (0-1) using hourly temperature, relative humidity, wetness, and hours of darkness. Biological parameters used to calculate these factors were obtained from Blaeser and Weltzien (1978; 1979), and Leu and Wu (1982). CYCLE predicted infection periods only when environmental conditions were favorable for infection and viable sporangia were available. Finally, the risk of infection (0-100) was calculated by multiplying the factors of spore survival and environmental favorability.

3. Methods

3.1 Weather and disease data.

Daily maximum and minimum temperatures and rainfall from 21 September of the previous year and hourly temperature, relative humidity, and wetness periods during the vine growing season are needed as inputs to DMCAST. In this study, daily data from 1985-1992 were obtained from the Climatological Reference Station located on the Vegetable Crop Research Farm of the New York State Agricultural Experiment Station (NYSAES), Geneva, NY and hourly data from the years 1985-1989 were digitized (Seem *et al* 1979) from hygrothermograph and surface wetness recorder charts which were collected from the Climate Reference Station and nearby research orchards of NYSAES. In 1990, hourly weather data were measured and collected using a micrologger (ENVIROCASTER, Neogen Co., Lansing, MI 48912) to which an aspirated wet bulb/dry bulb temperature sensor, surface wetness sensor, and tipping-bucket rain gauge were connected. The weather monitoring system was installed in an experimental vineyard at the Robbins Farm of NYSAES, about 1.5 km from the Climate Reference Station.

The vineyard was planted to the *Vitis* interspecific hybrid cultivar Chancellor, which is extremely susceptible to downy mildew. Observations of secondary cycles of downy mildew development were not made regularly during the growing seasons of 1985 through 1989. However, incidence and severity of downy mildew in the Chancellor vineyard was recorded from 1985-1989 on unsprayed vines used as controls in fungicide efficacy trials (Pearson and Riegel 1987; Pearson and Riegel 1992; Pearson, unpublished data). Field notes from daily to weekly inspections of the vineyard were also available (Pearson, unpublished). These data were used to identify the first disease occurrence dates and the levels of disease incidence severity during the five growing seasons in Geneva, New York.

In 1990-1993, downy mildew development on Chancellor grapevines was monitored throughout the growing season at the Robbins Farm. The date of first symptoms of downy mildew, as indicated by a distinctive epinasty of infected rachises, was recorded. Three four-vine panels of unsprayed grapevines in a randomized complete block design within the Chancellor vineyard were examined at 1-2 day intervals, beginning at 2.5 cm of shoot growth until the first symptom occurrence. The percentage of diseased clusters per vine, and the severity of disease on infected clusters was recorded at 7-10 day intervals after initial occurrence.

3.2 Dynamics of oospore maturation.

Infected leaves containing oospores of *P. viticola* were collected on 28 September, 1989 from the Chancellor vineyard. After confirming presence of many oospores in the infected leaf tissue under a microscope, leaf disks of 6 mm in diameter were cut using a cork-borer. The disks were sealed in 12 X 40 mm tubes made of plaster of Paris as described by Ronzon-Tran Manh Sung and Clerjeau (1988). Tubes containing leaf disks were buried at 5 cm below the soil surface in two vineyards at Geneva, and Riverhead, NY.

Leaf disks were examined seven times at 15-20 day intervals beginning 14 March 1990. From 200 to 500 oospores were removed from each set of leaf disks, placed on 1% water agar plates, and incubated at 20° C. The frequency of germinated oospores was assessed daily with a microscope (Ronzon-Tran Manh Sung and Clerjeau 1988). Frequencies of oospore germination observed over the sampling time period in both Geneva and Riverhead were fitted to the normal distribution density function. Parameters of the function were estimated by the Marquardt method of the NLIN procedure of the SAS system (SAS Institute 1985). The frequency distribution estimated from the observed data was compared with the expected frequency of oospore maturation predicted by DMCAST.

3.3 Primary infection.

To validate the primary infection model of DMCAST, the dates of primary infection and initial sporulation predicted by DMCAST were compared to the dates on which symptoms appeared in 1985-1993. The expected date of initial sporulation for each growing season was determined by adding number of days needed to pass the latent period after the expected dates of primary infection.

3.4 Secondary infection.

The secondary infection cycle model also was evaluated based on the levels of downy mildew severity observed in 1985-1993. Disease severity on Chancellor fruit clusters was correlated to: the number of discrete infections periods determined by DMCAST; the sum of the length of infection periods; and the sum of the mean risk value (calculated by DMCAST) of each infection period. In 1990, 1992, and 1993 the stepwise increase of disease incidence and severity on fruit from one assessment date to the next for unsprayed Chancellor vines was compared by correlation to the occurrence and severity of infection periods predicted by DMCAST.

3.5 Timing of fungicide applications according to model predictions.

One of the DMCAST model outputs was an hourly risk value for an infection period. A risk value was the product of relative, non-dimensional model estimates of spore survival and environmental favorability for infection, and varied between 0 and 100. A non-zero risk value during any hour indicated that an infection period was predicted, while the magnitude of the risk value was one indicator of the severity of the infection period. We applied the fungicide metalaxyl (Ridomil 2E, Ciba-Geigy Corp.) to Chancellor grapevines in 1992 and 1993 following infection periods predicted by DMCAST. Three post-infection schedules and one protectant schedule were devised: sprays were applied within 96 h after infection periods in which the risk value exceeded a threshold of (i) 50, (ii) 75, or (iii) 90 at any time during the infection period, and (iv) vines were also treated with protectant applications of metalaxyl at 14 day intervals. Based upon earlier trials (Pearson and Riegel 1987), a fungicide spray was assumed to protect the vines from subsequent infection periods for 14 days after application. Therefore, no action was taken if an infection period occurred within 14 days after a fungicide application in the post-infection treatments. Metalaxyl was applied at 2.33 L/ha (concentration of 1.245 mL of formulated product/L of spray volume) to four-vine plots arranged in a randomized complete block design, replicated four times. Sprays were applied with a hooded-boom sprayer at 2069 kPa and 1871 L/ha. DMCAST treatments were applied in 1992 following infection periods on 23 June (Risk level 50 and 75), 10 July (Risk level 50), 16 July (Risk level 75 and 90), and 30 July (Risk level 50). Protectant sprays were applied on 19 and 28 May, 11 and 25 June, 9 and 24 July, and 6 August 1992. DMCAST treatments were applied in 1993 following infection periods on 22 June (50, 75, and 90 Risk levels), 18 July (Risk level 50), and 22 July (Risk level 75). Protectant sprays were applied on 21 May, 4 and 17 June, 1 and 16 July, and 1 August 1993. Fifty fruit clusters from the center two vines of the four-vine plots were rated for disease on the Barratt-Horsfall scale and were converted to percentages of the cluster surface infected using Elanco conversion tables (cited in Kranz 1988). Treatment means were compared by the Waller-Duncan k-ratio t-test at p = 0.05.

4. Results

4.1 Oospore maturation.

Oospore matured earlier in Riverhead than in Geneva (Fig. 2). DMCAST also predicted earlier maturation of oospores in Riverhead than in Geneva. The normal distribution density functions which were fitted to the observed data indicated approximately 25-day difference in oospore maturation between Riverhead and Geneva.

When compared with the normal distribution density function which was fitted to the observed data for Geneva, prediction by the oospore maturation model of DMCAST appeared to be very accurate; the normal density curves of predicted and observed oospore germination were nearly identical (Fig. 2). However, oospores in the samples from Riverhead matured approximately 20 days later than the prediction by DMCAST. The standard deviation of the observed oospore maturation dates for Riverhead was less than predicted standard deviation (Fig. 2).

4.2 Primary infection.

The observed dates of symptom appearance (Table 1) indicate that downy mildew was present by the time of the assessment, but do not indicate that the disease was completely absent before the date of assessment. Therefore, the data can only be used to partially validate our model: DMCAST should not predict a date of primary infection later than the date of first symptoms. In all years except 1987 and 1990, the first symptoms of downy mildew were observed within two weeks after DMCAST predicted primary infection (Table 1). In 1987 and 1990, symptoms of downy mildew were not observed until 26 and 18 days, respectively, after DMCAST predicted primary infection (Table 1).

DMCAST predicted initial sporulation 6-13 days after the primary infection in all years except 1988 (Table 1). In 1988, DMCAST predicted sporulation 40 days after the primary infection. The initial sporulation dates determined by DMCAST for 1985, 1986, and 1989-1992 were within one week of the first dates of disease observation. For 1987 and 1988, the expected dates of initial sporulation were 15 days prior to and 26 days after the first observation of disease, respectively.

Average monthly temperatures during May and June of 1985-1992 were neither abnormally warm nor cold, and were within one standard deviation of the 30-year mean for 1960-1989. However, rainfall during May and June in 1988 was far below normal, resulting in conditions too dry for downy mildew development. May and June of 1991 were also warmer and drier than all other years of the study except 1988.

4.3 Secondary infection cycles

The length of latent period after the primary infection varied from year to year. Environmental favorability values indicated that favorable conditions for infection occurred even before the primary infection in the spring of all years. A larger number of infection periods, and a resultant greater number of hours of infection and cumulative severity of infection was noted for the years 1985-1989 as compared to 1990-1993 (Table 2).

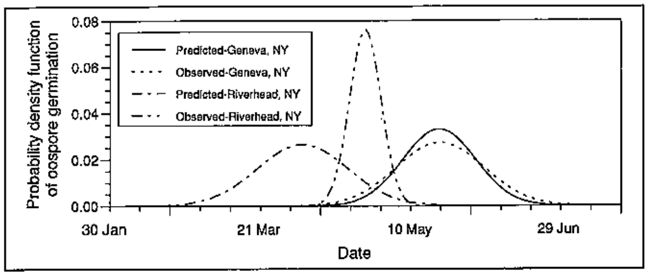

Fig. 2: Calculated frequency distribution of oospore germination based on field observations and predictions made by the OSP model of DMCAST for 1990 at two locations.

Abb. 2: Berechnete Häufigkeitsverteilung der Oosporenkeimung basierend auf Feldmessungen. Die Vorhersagen wurden mit dem OSP-Modell von DMCAST für 1990 für zwei Standorte vorgenommen

Fig. 2: Distribution des frequences calculees de germination des oospores basee sur les observations sur le terrain et les previsions faites le modele OSP de DMCAST for 1990 en deux sites differents.

Tab. 1: Observed and expected initial development of grape downy mildew on vines of the *Vitis* interspecific hybrid cultivar Chancellor in Geneva during the growing seasons of 1985 through 1992.

Tab. 1: Beobachtete und erwartete Anfangsentwicklung des Falschen Mehltaus an der Hybride "Chancellor" in Geneva während der Vegetationszeiten von 1985 bis 1992

Tab. 1: Développement initial observé et attendu du mildiou de la grappe chez l'hybride Chancellor à Genova pendant les périodes végétatives de 1985 à 1992.

Year	Observed appearance of symptoms	Phenology at symptom appearance	Expected date by DMCAST	
			Primary infections	Initial sporulation
1985	18 June	Bloom	6 June	16 June
1986	6 June	Bloom	24 May	1 June
1987	23 June	10 days postbloom	28 May	8 June
1988	16 June	1 day prebloom	2 June	12 July
1989	14 June	7 days prebloom	10 June	21 June
1990	14 June	6 days prebloom	27 May	9 June
1991	4 June	3 days prebloom	24 May	30 May
1992	14 June	7 days prebloom	7 June	17 June
1993	21 June	Bloom	14 June	20 June

Table 2: Severity of grape downy mildew on fruit clusters of the *Vitis* interspecific hybrid cultivar Chancellor and the number of infection periods, accumulated hours of infection, and mean accumulated risk levels of infection periods determined by DMCAST in Geneva during the growing seasons of 1985 through 1993.

Tab. 2: Befallsstärke des Falschen Mehltaus an Gescheinen an der Hybride Chancellor und die Anzahl der Infektionsperioden, summierte Infektionsstunden sowie das mittlere summierte Risiko der Infektionsperioden bestimmt mit DMCAST in Geneva während der Vegetationsperioden 1985 bis 1992

Tab. 2: Sévérité des attaques de mildiou de la grappe sur l'Hybride „Chancellor" et nombre de périodes d'infection, nombre d'heures d'infection cumulées, et valeur moyenne des risques cumulés des périodes d'infection, déterminés par DMCAST à Genova pendant les périodes végétatives de 1985 à 1992.

Year	Severity[w] of disease on fruit clusters	Number[x] of infection periods[b]	Accumulated[y] hours of infection	Accumulated[z] risk of infection
1985	13.2	29	221	9870
1986	83.7	83	580	31981
1987	7.7	50	368	25172
1988	1.0	31	274	16177
1989	96.7	34	170	10809
1990	27.2	21	146	9344
1991	4.1	9	71	3468
1992	50.4	18	136	6867
1993	8.7	7	47	3038

[w]Ratings were made on 8 Aug 1985, 14 Jul 1986, 12 Aug 1987, 11 Jul 1989, 9 Aug 1990, 21 Aug 1991, 21 Aug 1992, and 3 Aug 1993. nd = no data.

[x]Number of discrete infection periods determined by DMCAST for the growing season.

[y]Summation of the duration of infection periods, in hours.

[z]Summation of the mean infection risk level of each infection period during the growing season. The mean risk value of each period was determined by dividing the sum of the hourly risk values by the number of hours in the infection period.

Fewer infection periods occurred in 1990-1992 than in 1985-1989 (Table 2). Simple linear regressions of disease severity on fruit were attempted against the following independent variables: (i) total number of infection periods, (ii) number of infection periods by 1 July, (iii) number of infection periods during July, (iv) number of infection periods during August, (v) number of infection periods during June and July, and (vi) number of infection periods during July and August. Similar analyses were attempted for the total hours of infection, and the summation of mean risk values for infection periods during the above intervals. Coefficients of correlation in analyses i-vi above were 0.32, 0.22, 0.12, 0.44, 0.21, and 0.30, respectively. Corresponding coefficient values for hours of infection were 0.17, 0.14, 0.01, 0.35, 0.07, and 0.14, respectively, while those for the summation of mean risk values were 0.14, 0.09, 0.01, 0.40, 0.05, and 0.14. With $n = 9$ and $df = 8$, none of the coefficients of correlation were significant at $p = 0.05$. Because there were only nine observations (Table 2), and because of the high degree of colinearity among the above variables, multiple regressions were not attempted. Over a 3-yr period, disease incidence increased significantly following most infection periods predicted by DMCAST (Table 4). However, there was no increase in disease incidence after 1 of 11, 6 of 19, and 1 of 5 infection periods in 1990, 1992, and 1993, respectively (Table 4). In no case did disease increase in the absence of infection periods predicted by DMCAST, *i.e.*, when conditions were predicted to be unfavorable for epidemic development (Table 4).

4.4 Timing of fungicide applications according to model predictions

Metalaxyl applied in a protectant program of 6-7 sprays provided nearly complete control of downy mildew on Chancellor fruit in 1992 and 1993 (Table 3). Disease severity was higher on unsprayed vines in 1992 than in 1993 (Table 3). Although the amount of infection that resulted from the post-infection sprays applied after infection periods in 1992 was above what would be tolerated in most commercial vineyards, disease was nonetheless reduced from 50.4% to 12.6% by a single post-infection fungicide application (Table 3). In 1993, a single post-infection application provided a reduction of disease that was equal to that provided by six protectant applications.

Ridomil 2E was applied at 2.33 L/ha (concentration of 1.245 ml of formulated product/L of spray volume) to four-vine plots arranged in a randomized complete block design, replicated four times. Sprays were applied with a hooded-boom sprayer at 2069 kPa and 1871 L/ha. DMCAST treatments were applied in 1992 following infection periods on 23 June (Risk level 50 and 75), 10 July (Risk level 50), 16 July (Risk level 75 and 90), and 30 July (Risk level 50). Protectant sprays were applied on 19 and 28 May, 11 and 25 June, 9 and 24 July, and 6 August 1992. DMCAST treatments were applied in 1993 following infection periods on 22 June (50, 75, and 90 Risk levels), 18 July (Risk level 50), and 22 July (Risk level 75). Protectant sprays were applied on 21 May, 4 and 17 June, 1 and 16 July, and 1 August 1993. Treatment means within a year that are followed by the same letter are not significantly different according to the Waller-Duncan k-ratio t-test ($p \leq 0.05$).

Tab. 3: Severity of downy mildew on fruit clusters of the *Vitis* interspecific hybrid cultivar Chancellor in 1992 and 1993. Fungicide sprays were timed according to DMCAST infection risk levels or were applied as protectants at approximately 14-day intervals.

Tab. 3: Stärke des Mehltaubefalls an Trauben der Hybride "Chancellor" während der Jahre 1992 und 1993. Die Fungizidbehandlungen wurden entsprechend den Berechnungen des DMCAST-Modells festgelegt oder wurden protektiv ungefähr alle 14 Tage appliziert.

Tab. 3: Sévérité du mildiou sur les grappes de l'hybride „Chancellor" en 1992 et 1993. Les applications de fongicides sont définies en fonction des degrés de risque d'infection selon DMCAST ou faites en tant que protections à approximativement 15 jours d'intervalles.

Year	Treatment	Number of sprays	Severity of infection on fruit clusters
1992	Protectant	7	0.1 a
	50	3	9.3 b
	75	2	9.4 b
	90	1	12.6 c
	Control	0	50.4 d
1993	Protectant	6	0.0 a
	50	2	0.1 a
	75	2	0.4 a
	90	1	0.1 a
	Control	0	8.7 b

5. Discussion

DMCAST was able to simulate the dynamics of oospore maturation of *P. viticola* in Geneva. However, the rainfall parameters used in the oospore maturation model needed to be adjusted depending upon

Tab. 4: Change in the incidence of infection of fruit clusters of the *Vitis* interspecific hybrid cultivar Chancellor between disease assessments, and occurrence of downy mildew infection periods according to DMCAST.

Tab. 4: Veränderungen in der Häufigkeit der Infektion von Trauben der Hybride "Chancellor" vom Krankheitsbefall bis zum Auftreten des Falschen Mehltaus entsprechend den DMCAST-Berechnungen

Tab. 4: Alterations de la fréquence d'infection des grappes de l'hybride „Chancellor" des symptômes de la maladie jusq'au début de mildiou selon les calculations de DMCAST

		Infection periods[x]		Relative change[y] in disease	
Year	**Interval**[w]	**Duration(s) (h)**	**Risk level(s) (%)**	**Incidence**	**Severity**
1990	27 May - 14 Jun	6	55	(0.64%)*	ND
	14 Jun - 21 Jun	3	28	NS	ND
	21 Jun - 29 Jun	9	70	4.3*	ND
	29 Jun - 9 Jul	7,17	90,78	4.3*	ND
	9 Jul - 17 Jul	4,5	69,60	2.9*	ND
	17 Jul - 24 Jul	3,8,	18,94	1.5*	ND
	24 Jul - 1 Aug	9,17	76,82	1.3*	ND
1992	2 Jun - 14 Jun	0	0	(0.1%)*	(0.5%)*
	14 Jun - 14 Jul	3,35,11,9,3,4,4,1	2,82,53,44,15,42,14,71	42.0*	16.6*
	14 Jul - 21 Jul	1,4,20,9	18,82,94,43	NSNS	
	21 Jul - 28 Jul	7,6	66,49	NS	NS
	28 Jul - 6 Aug	2,1,7	65,32,18	2.3*	NS
	6 Aug - 20 Aug	6,3,	49,15	3.2*	2.0*
1993	14 Jun - 21 Jun	0	0	(0.1%)*	(0.5%)*
	21 Jun - 28 Jun	9,8	69,94	16.0*	NS
	28 Jun - 7 Jul	5	22	NS	NS
	7 Jul - 15 Jul	0	0	NS	NS
	15 Jul - 15 Aug	6,14	82,73	2.5*	NS

[w]Interval between two assessments of disease incidence and severity on 10-38 fruit clusters/vine. The percentage of infected fruit clusters (incidence) and the mean percentage of the cluster surface infected (severity) was recorded on unsprayed vines in four-vine plots replicated four times.

[x]Infection periods forecasted by DMCAST that could have resulted in observed changes in disease incidence and severity during the interval between two disease assessments, given a minimum interval of 5 days between infection and symptom expression. Duration of each infection period was the number of hours during which the risk of infection was > 0. Risk level indicates the maximum risk level reached during respective infection periods.

[y]Relative change in disease computed by dividing disease incidence or severity at the end of the assessment interval by disease at the start of the interval. Values in parentheses indicate disease incidence or severity when disease was first found during a growing season. An asterisk indicates a significant change from the previous assessment according to Student's t-test at p = 0.05. ND indicates no data. NS indicates that no significant change occurred during the interval.

locations for which the model was applied. The failure to describe accurately the dynamics of oospore maturation in Riverhead may have been due to the parameters, which were calculated from rainfall data measured in Geneva from 1967 through 1986. The oospore maturation model of DMCAST was adapted from a model (POM) which was based on the assumption that the fungus adapts to local climate conditions (Strizyk 1983).

DMCAST differed from the POM model in handling the mature oospore population that could contribute primary infections in the spring. The POM model calculates a date when the distribution of oospores reaches a maximum (DOM) based on the daily rainfall beginning in September, and predicts severe development of downy mildew in the spring when DOM occurs early. In DMCAST, it was assumed that frequency of mature oospores distributed normally over time, and that individual cohorts of oospores accumulated under the influence of environmental conditions. The estimated frequency of matured oospores was accumulated daily until this exceeded 3%. It was then considered that enough inoculum existed for primary infection. However, we required an additional condition to be satisfied before primary infection was predicted: vine growth at or beyond the Eichhorn-Lorenz stage 12. This condition eliminated early predictions when oospores were estimated to be mature but vines were not susceptible to infection.

The DMCAST prediction of the date of primary infection appeared to be accurate when compared with the first observed date of downy mildew symptoms on Chancellor, a highly susceptible cultivar (Table 1). Rachises of Chancellor display a distinctive epinasty within 4-7 days following infection by *P. viticola*, and can thus be recognized even before sporulation occurs. Symptoms of downy mildew infection were observed on Chancellor clusters 4-25 days after primary infection was predicted by DMCAST (Table 1), and never before the predicted date of primary infection. From 1985-89, when 7 days often elapsed between vineyard assessments, symptoms were observed 12, 13, 25, 14, and 4 days, respectively, after the predicted date of primary infection. From 1990-93, when the frequency of vineyard assessments was increased to every 1-2 days, symptoms were observed 18, 11, 7, and 7 days, respectively, after the predicted date of primary infections. Substracting a 4-7 day latent period from the above values would indicate a maximum error of 0-18 days for the period 1985-89, and 3-11 days for the years 1990-93. Considering the interval between disease assessments is a source of error not attributable to DMCAST, the model was judged sufficiently accurate in predicting the date of primary infection.

In DMCAST, the incubation period was calculated only once after the date of primary infection. Since the incubation period was affected by temperature (Muller and Rabanus 1923), the yearly variation in number of days between the dates of primary infection and initial sporulation (Table 1) was due to differences in temperature after primary infection. After initial sporulation, it is difficult to presume that spores of each generation would go through infection cycles simultaneously. Therefore, incubation period was no longer considered in the secondary infection cycle model of DMCAST.

The lack of significant correlation between numbers of infection periods, and summation of hours of infection or severity of infection highlight the difficulties in anticipating the severity of a particular year for downy mildew (Table 2). Years with relatively low numbers of infections periods, such as 1992, can produce severe fruit infection; while years such as 1987, in which the number of infection periods, the hours of infection, and the mean severity of infection periods were all relatively high, result in low levels of disease. Some of this variation might result from year-to-year variation in primary inoculum dose. However, in 1991, 44% of the fruit clusters on unsprayed Chancellor vines aborted due to the first cycle of primary infection, a fact not reflected by the low level of disease observed on the surviving fruit clusters later in the year. There are additional sources of variation in disease severity that are not explained by variations in either primary inoculum, or simple summations of numbers, severity, or length of infection periods. Factors such as the role of rainfall in dispersal of sporangia and zoospores, and acquisition of ontogenic resistance in grapevine are poorly understood, but potentially important epidemiological factors. Neither factor is presently a component of DMCAST or other models of grape downy mildew. Surface wetness and high relative humidity are required for infection, but infection periods caused by dew are not distinguished from those initiated or prolonged by rain. The potential for splash dispersal of inoculum, the jarring effect of rain on detachment of sporangia, and the turbulent air associated with rain event contrasts sharply to the predominant environment of a dew period: little or no splash dispersal, leaves not subjected to impact of rain droplets, and relatively still air. Although rain occurred immediately before or during all but two of the 46 infection periods in 1990, 1992, and 1993, the intensity and distribution of rain varied greatly (Table 2). Finally, variation in results may be caused by disparities associated with the representation of surface wetness on leaves by most artificial surface wetness sensors. *Plasmopara viticola* infects through stomata, which are hypophyllous in most *Vitis* species, and yet dew forms initially, and often exclusively on the upper leaf surface. Electronic grid sensors, as they are normally deployed detect surface wetness from either rain or dew on the upper surface of the sensor. How this relates to surface wetness at the infection court for downy mildew is also poorly understood.

During each infection period, DMCAST calculated the proportion of the a population of spores that is likely to survive to cause infection. These values range from 0-1. No attempt was made to model the absolute increase of disease, nor was the above proportion weighted to account for an ever-increasing pool of inoculum after each cycle of infection. Of all coefficients of correlation obtained when disease severity was regressed against the number of infection periods, the hours of infection, or the summation of the mean risk values of infection periods, the highest correlations were obtained when the independent variables were restricted to August (0.44, 0.35, and 0.40 for the number of infection periods, the hours of infection, and the summation of the mean risk values of infection periods, respectively). The progressive increase in diseased tissue and the quantity of secondary inoculum in late-season infection periods may be reflected in these results.

Despite the above described limitations of both our model and environmental sensors, infection periods predicted by DMCAST in 1990, 1992, and 1993 were frequently associated with a stepwise increase of disease incidence and severity on fruit clusters of Chancellor grapevines (Table 4). Again, no clear association was apparent between the number, length, or severity of infection periods and the subsequent increase of disease between assessment dates (Table 4). Four consecutive infection periods prior to the assessments performed on 14 and 21 July 1992 did not result in a significant increase in disease incidence or severity. However, disease incidence increased 4.3-fold, 4.3-fold, and 2.9-fold, respectively, following 1, 2, and 2 infection periods in 1990 (Table 4). Most importantly, disease did not increase when DMCAST predicted that conditions were unfavorable for downy mildew (Table 4). When an error is made in predicting the occurrence of downy mildew, DMCAST consistently overestimates the threat of disease.

In our disease control trials, an infection risk threshold level of 50, 75, or 90 calculated by DMCAST was used to trigger a spray recommendation. Our assumption was that the maximum risk reached during any hour of the infection period would provide an indication of the actual increase of disease that would result from the infection period. The decision to use the maximum hourly risk value of an infection period was arbitrary, and was made before the relationship between DMCAST predictions and disease development was completed. Other possible forms of the model output that could have been used for the same purpose would include: (i) the mean risk level of an infection period, (ii) the length of the infection period, and (iii) some product of risk and length. Given the poor correlation between the risk level of infection periods and the subsequent increase of disease (Table 4), a more logical approach for future studies would be to apply a post-infection spray when the risk of in-fection exceeded 0. A non-zero risk value indicates that an infection period has occurred. Considering the uncertainty surrounding the validity of DMCAST ratings of infection period severity, and the lack of correlation between the infection period length and disease development, this may be the most prudent use of the model predictions. DMCAST may not accurately describe the severity of infection periods, but it does provides a usable prediction of the occurrence of both primary and secondary infection. The use of a non-zero threshold value would have resulted in the application of four and three applications of metalaxyl in 1992 and 1993, respectively. We are currently developing a program for the simultaneous management of the major fungal diseases of grapevine in the Northeastern US (Gadoury 1993): powdery mildew (*Uncinula necator*) downy mildew (*Plasmopara viticola*), black rot (*Guignardia bidwellii*), and Phomopsis cane and leaf spot and fruit rot (*Phomopsis viticola*). For cultivars of *Vitis vinifera* and disease-susceptible interspecific hybrid cultivars, the growing season is divided into three phases. In the first, or prebloom phase, post infection applications of myclobutanil or triadimefon are directed against rain-initiated infection periods for powdery mildew and black rot. The second, or transition phase, occurs near bloom. For approximately two weeks, primary inoculum of powdery mildew, black rot, and downy mildew are present together, and the three diseases should be managed as a complex of diseases. Neither myclobutanil or triadimefon control downy mildew. The utility of DMCAST in this management program, is that it can accurately predict when the inclusion of a material for downy mildew is required during the transition phase. If an infection period for downy mildew occurs simultaneously with a rain-initiated infection period for either black rot or powdery mildew, then a fungicide with post-infection activity against downy mildew can be used in combination with either myclobutanil or triadimefon. If an infection period for powdery mildew or black rot occurs after DMCAST indicates that the 3% threshold of oospore maturity has been reached, but before primary infection has occurred, then a protectant fungicide for downy mildew, such as mancozeb, can be used in combination with myclobutanil or triadimefon. In the third and final phase of the management program (post-bloom), the supply of primary inoculum for powdery mildew is exhausted, and black rot fruit infection is unlikely if primary infection has been successfully controlled through the transition phase. The emphasis of the program then changes to post-infection control of downy mildew. DMCAST can then be used to time post-infection fungicide applications for continued suppression of downy mildew.

Summary

DMCAST is a weather-driven prediction model that simulated primary and secondary infection of grapevine by the downy mildew pathogen (*Plasmopara viticola*). Oospores maturity was estimated from precipitation during overwintering. A forecast of primary infection required cumulative oospore maturity > 3%, and > 5 leaves/shoot. After primary infection occurred, DMCAST predicted the occurrence and severity of secondary infection periods based on temperature, relative humidity, leaf wetness, and hours of darkness. DMCAST accurately predicted the date of initial disease for 9 consecutive years. Over 3 years, disease increased significantly following most, but not all, forecasted periods of secondary infection. Disease never increased in the absence of forecasted infection periods. DMCAST is an accurate predictor of primary infection, and an imperfect but useful predictor of secondary infection periods.

Zusammenfassung

DMCAST stellt ein auf Wetterdaten basierendes Modell dar, das primäre und sekundäre Infektionen von Weinreben durch falschen

Mehltau (*Plasmopara viticola*) simuliert. Die Vorhersage des Oosporenreifegrades basierte auf der Niederschlagsmenge über die Wintermonate. Wahrscheinliche Primärinfektionen erforderten einen kumulierten Oosporenreifegrad >3% und mehr als 5 ausgebildete Blätter pro Rebtrieb. Nach erfolgter Primärinfektion basierte die Vorhersage des Zeitpunktes und der Schwere von sekundären Infektionen durch DMCAST auf Temperaturdaten, Luftfeuchtigkeit, Dauer von Blattfeuchteperioden und Dauer von Dunkelperioden. DMCAST Vorhersagen von Primärinfektionen waren in 9 aufeinanderfolgenden Jahren präzise. Über einen Zeitraum von 3 Jahre folgte der Krankheitsverlauf in den meisten aber nicht in allen Fällen den vorhergesagten Perioden von Sekundärinfektionen. Der Grad der Erkrankung erhöhte sich allerdings nie, wenn Sekundärinfektionen nicht vorhergesagt wurden. DMCAST stellt sich dar als ein zuverlässiges Modell zur Vorhersage von primären Infektionen, und als verbesserungswürdiges aber inzwischen brauchbares Instrument zur Vorhersage von Sekundärinfektionen.

Résumé

DMCAST est un système de prédiction qui simule les infections primaires et secondaires du mildiou (*Plasmopara viticola*) de la vigne en se basant sur les données météorologiques. La maturité des oospores a été estimée à partir des précipitations d'hiver. Les prévisions d'infections primaires ont nécessité simultanément une maturité d'oospore supérieure à 3% et au moins 5 feuilles par rameau. Après les infections primaires, DMCAST a prédit le développement et la sévérité des périodes d'infections secondaires d'après les températures, l'hygrométrie de l'air ambiant, l'humidité des feuilles et les heures d'obscurité. Durant 3 ans, le mildiou a progressé significativement en suivant la plupart mais pas toutes les périodes prédites d'infections secondaires. La maladie n'a jamais progressé en dehors des périodes prédites d'infections. DMCAST est un prédicateur précis des infections primaires et un prédicateur utile mais imparfait des périodes d'infections secondaires.

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